KONSTANTINOS MITSIDES

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Research Interests

Scalable exploration and adaptive learning for open-ended discovery in AI: reinforcement learning, LLM-driven evolutionary methods, unsupervised environment design, and quality-diversity.

Publications

K. Mitsides, M. Faldor, A. Cully – Dreaming in Code for Curriculum Learning in Open-Ended Worlds at ICML 2026

K. Mitsides, M. Faldor, A. Cully – Scaling PG-QD with Massive Parallelization via Behavioral Variations at GECCO 2025

Research Experience

PhD Researcher, British Open-Ended Learning & Discovery (BOLD) Lab – Imperial College London **10/2024 - Present**

Supervisor: Prof. Antoine Cully · **BOLD:** EPSRC-funded national AI lab (Imperial · Oxford · UCL)

- **DiCode (ICML 2026):** Built an LLM-driven curriculum framework that generates executable environment code to scaffold RL agents in open-ended worlds, achieving 17% higher mean return than the strongest baseline and unlocking late-game skills where prior methods score 0%.
- Engineered a single vmapped, JIT-compiled PPO loop running 1000+ parallel environments that each execute different LLM-generated task code, supporting heterogeneous per-environment logic under one vmap.
- Built an asynchronous environment-generation pipeline overlapping 235B-parameter LLM inference with RL training, cutting training-blocking inference >12×. **Stack:** JAX, Craftax, PPO + Gated Transformer-XL, Qwen3-235B.
- **ASCII-ME (GECCO 2025):** Developed the first policy-gradient quality-diversity algorithm without centralised actor-critic training, producing a diverse collection of high-performing deep-neural-network policies in <250s on a single GPU.
- Ran on average 5× faster than state-of-the-art while achieving +25% QD score at an equal evaluation budget on locomotion benchmarks. **Stack:** JAX, Brax, QDax, MAP-Elites, policy gradients.

Education

PhD Artificial Intelligence, Imperial College London **10/2024 - Present**

Scholarships: EPSRC DTP Studentship; Scholarship Foundation of the Cyprus Government

MSc Artificial Intelligence, Imperial College London **10/2023 - 10/2024**

Final Grade: Distinction – "Distinguished" status for MSc thesis (87%)

BSc Mathematics, University College London (UCL) **09/2020 - 09/2023**

Final Grade: First Class Honours

Scholarships: Academic Excellence Scholarship – Scholarship Foundation of the Cyprus Government

Relevant Courses: Probability & Statistics, Stochastic Processes, Algebra, Mathematical Methods, Analysis, Graph Theory

The English School, Nicosia, Cyprus **09/2012 - 06/2019**

A-Levels: Further Mathematics (A*), Mathematics (A*), Physics (A*), Greek (A) – Highest Marks in Europe & Cyprus

Technical Skills

Programming & ML: Python, JAX, PyTorch, Hugging Face, vLLM

Infrastructure & Tooling: Git, Linux, Docker, Apptainer, Weights & Biases (WandB)

Experience & Competitions

Graduate Teaching Assistant – Imperial College London **01/2025 - Present**

- Delivered a guest lecture on deep learning in Python; mentored students through end-to-end ML projects.
- Co-supervised an MSc thesis in reinforcement learning and evolutionary computation, which achieved Distinction.
- Led tutorials and graded assignments across deep learning, natural language processing, and Python modules.

The Data Open Europe 2021 / Ready Trader Go – Citadel / Optiver **10/2021 - 11/2021**

- Performed sentiment analysis, uncovering higher engagement for headlines with negatively charged content.
- Developed a profitable trading bot to enhance order-book liquidity, leveraging historical data analysis.

Staff Sergeant – National Guard of Cyprus **07/2019 - 09/2020**

- Led and scheduled a team of 30 soldiers, making rapid decisions to resolve unexpected operational challenges.

Chief Engineer – F1 in Schools World Finals **09/2017 - 09/2018**

- Designed and produced a miniature CAD car (SolidWorks), reaching the World Finals as Cyprus' second-fastest.

Additional Information

Academic Service: Invited Reviewer, NeurIPS 2026 (Top Reviewer)

Talks & Panels: Panellist on the future of AGI, 'The Thinking Game' screening (Made by Many, London, 2025)

Maths Competitions: Top-3 placings in National Maths Olympiads & Kangaroo Maths Competitions

Sports / Music: National records & multiple medals in swimming; Grade 8 Piano & Theory of Music

Languages: English (fluent), Greek (native)